By Satvik Pandey

Approach Document
SHL Conversational Assessment Recommender

This document summarises my implementation of a stateless FastAPI conversational recommender for SHL Individual Test Solutions. The service accepts full chat history per request and returns a grounded shortlist (1-10 assessments) with validated catalog URLs only.

1) Design choices

The key decision was to use a retrieval-grounded conversational pipeline instead of direct prompting over the full catalog. User intent is often incomplete in turn 1, so the agent must clarify before recommending. I added explicit policy guards for scope control (legal/refusal) and end-of-conversation handling to keep responses consistent under an 8-turn budget.

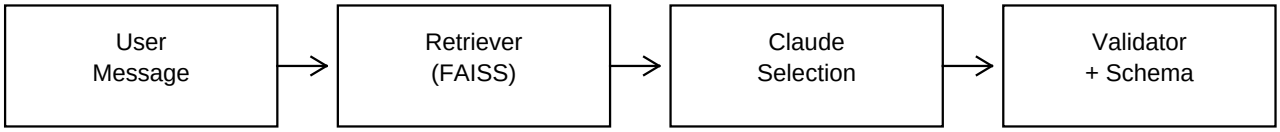
2) Catalog preparation

The raw shl_product_catalog.json contains malformed embedded newlines in string values, so a pre-parse sanitiser was required. I filtered out pre-packaged bundles (* Solution, * Short Form, Job Focused Assessment) to enforce assignment scope. The final catalog has 370 individual tests. test_type is derived deterministically from catalog keys to guarantee stable API formatting.

3) Retrieval setup

I used sentence-transformers/all-MiniLM-L6-v2 + FAISS inner-product search, with keyword overlap boosting and query expansion (e.g., leadership, sales, Java, contact center). This narrows each turn to top-25 candidates, reducing latency and hallucination risk. To fit Render free-tier memory, the index and model cache are pre-built during Docker build and loaded at runtime.

System flow (runtime request path)



Only top candidates are passed to the LLM; every returned URL is whitelisted against the local catalog.

4) Prompt design and policy orchestration

The system prompt forces JSON output and catalog grounding. I run rule checks before the LLM call: (a) vague first-turn -> one clarifying question, (b) legal/compliance -> refusal, (c) compare mode uses catalog descriptions only. Claude Sonnet then selects URLs from candidate items, and a hard validator drops any URL outside the whitelist. This provides deterministic safety on top of non-deterministic generation.

5) Evaluation method

I parsed gold URLs from C1-C10 and replayed full multi-turn traces against the live endpoint using eval/replay.py. Primary metric: Recall@10 on final shortlist. Secondary checks: schema compliance, catalog-only URL grounding, vague-turn clarification, and legal refusal behavior. Live benchmark: mean Recall@10 = 0.946 on C1-C10 with warm chat latency typically under 15 seconds.

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6) What did not work

Initial deployment failed on Render free tier due to memory pressure while loading sentence-transformers and building FAISS at startup. Also, relying purely on semantic retrieval produced lower recall on role-specific examples (e.g., graduate battery and full-stack refinement cases).

7) How I measured and improved quality

Improvements were made iteratively using replay traces and delta checks:
- Added deterministic guardrails (clarify/refuse/compare handling).
- Added targeted query expansion and domain-aware recommendation rules.
- Enforced strict URL whitelist and recommendation cap.
- Moved index/model build to Docker build stage and loaded cached artifacts at runtime.

Each change was verified with the same C1-C10 harness to confirm recall and behavior gains.

8) Deployment choices

Production uses Docker on Render with CPU-only PyTorch and pre-baked cache artifacts. This avoids runtime OOM and keeps startup predictable. Endpoints are intentionally minimal: /health and /chat only. A scheduled keep-warm job reduces free-tier cold-start impact during evaluation windows.

9) Stack and tools used

Python, FastAPI, Anthropic Claude Sonnet (claude-sonnet-4-20250514), sentence-transformers, FAISS, and custom evaluation harness scripts. Development iteration used AI-assisted coding support while keeping architecture, evaluation design, and final verification manual and test-driven.

10) Closing note

The final system is grounded, stateless, and aligned with assignment constraints: schema-safe responses, catalog-only recommendations, multi-turn refinement, and measurable evaluation quality. The deployed API is production-ready for SHL's automated replay checks.